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22 June 1978
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MILITARY SPECIFICATION

FILTER AIDS, WATER PURIFICATION

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable filter aid materials to be used in military designed water purification equipment incorporating pressure-type filtration for production of potable water.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-N-105	- Nails, Brads, Staples, and Spikes: Wire, Cut and Wrought.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).

FSC 9390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
UU-S-48	- Sacks, Shipping, Paper.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-P-15011	- Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th Street, NW, Washington, DC 20036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The filter aid shall consist of disposable filter material, such as milled calcined diatomaceous earth or milled expanded perlite.

3.2 Chemical composition. The filter aid shall have the following chemical composition:

Moisture, percent, maximum	0.75
Weight loss on ignition, percent, maximum	2.75
Silica (SiO_2), percent, minimum	64.00
Iron oxide (Fe_2O_3), percent, maximum	2.00
Aluminum oxide (Al_2O_3), percent, maximum	17.00
Magnesium oxide, percent, maximum	0.75
Other compounds, percent, maximum 1/	12.75

1/ Other compounds are lime, and oxides such as potassium monoxide, sodium monoxide, sulphur trioxide, and titanium dioxide.

3.3 Deleterious ingredients. When filter aid in the amount of 375 milligrams (mg) is added to one liter of water, 375 parts per million (ppm), the increase in the amount of the following deleterious ingredients in the water shall not exceed the amount specified:

Lead	0.025 mg/l
Copper	0.5 mg/l
Fluoride	0.80 mg/l
Arsenic	0.025 mg/l
Selenium	0.005 mg/l
Hexavalent chromium	0.025 mg/l

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Chloride	125.0 mg/l
Sulfate (SO_4)	125.0 mg/l
Zinc	2.5 mg/l
Phenols	0.0005 mg/l

3.4 Index value. The filter aid shall have an index valve not less than 10,400 nor more than 16,400 based on the following:

$$I = \frac{\gamma_F^{3/2} \beta_o^{1/2}}{\rho_F}$$

where I = index value
γ_F = bulk density
β_o = resistance index
ρ_F = effective specific gravity

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of all filter aid offered for acceptance at one time which has been produced by one manufacturer under the same manufacturing conditions, and with no change in materials, provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.4).

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4.3.2 Sampling. Sampling for tests shall be in accordance with MIL-STD-105. A specimen shall be removed from each bag in the sample and placed in a clean, dry container, labeled to identify the lot and the bag from which it was taken.

4.3.3 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.1 through 4.4.4.3.3. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Tests. Unless otherwise specified (see 6.2), distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections shall be applied where significant.

4.4.2 Chemical composition. The chemical composition of filter aid shall be determined in accordance with ASTM C 25. Nonconformance to 3.2 shall constitute failure of this test.

4.4.3 Deleterious ingredients. Place one gram of the specimen, weighed to the nearest mg, in a glass beaker and add enough double-distilled water to make a concentration of 375 ppm, plus or minus 25 ppm of filter aid. Stir for 6 hours with a magnetic stirrer at 100 revolutions per minute (rpm); cover and set aside for 16 hours with the temperature of the specimen kept at 27° C to 29° C. At the end of the 22 hour period, filter through filter paper to remove the undissolved constituents. Determine the amount of deleterious ingredients in accordance with Standard Methods for Examination of Water and Wastewater of the American Public Health Association. Nonconformance to 3.3 shall be cause for rejection.

4.4.4 Index value. The index value (I) shall be determined by use of Figure 1. Nonconformance to 3.4 shall be cause for rejection.

4.4.4.1 Apparatus. The following apparatus is required (see 6.3):

(a) Flask - A standard volumetric flask having a capacity of 500 milliliter (ml).
(b) Balance - A top loading balance with sensitivity and readability of 0.01 gm capable of accommodating the flask, and having a capacity of 1200 gm.

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(c) Thermometer - A mercury thermometer with a range of minus 20° C to 102° C graduated in 1.0° C subdivisions.
(d) Stirring apparatus - A magnetic stirrer complete with several plastic coated stirring bars and capable of variable speed of 100 to 150 rpm.
(e) Wash bottle.
(f) Air source - An air source capable of supplying air at a pressure not less than 40 pounds per square inch (psi). Air may be supplied from laboratory air line or compressor.
(g) Air pressure regulator - A pressure regulator capable of maintaining air at a pressure of 30 psi with an accuracy of plus or minus 0.1 psi.
(h) Pressure measuring device - A mercury manometer or pressure gage capable of measuring 30 psi, with an accuracy of plus or minus 0.3 psi at 30 psi.
(i) Filter cake support - Filter membranes having a pore size of 0.45 microns and 37 millimeters (mm) in diameter. Filter membranes are discarded after use and therefore a supply of 100 membranes should be available.
(j) Filtration apparatus - Filtration apparatus conforming to Figure 2. Sample holder shall conform to Figure 3 and filter cell shall conform to Figure 4.
(k) Absorbent pads - Absorbent pads for use with filter cake support (item 1), 37 mm in diameter. Absorbent pads are discarded with filter membranes after use and therefore a supply of 100 pads should be available.
(l) Graduated cylinder - A graduated cylinder having a capacity of not less than 1 liter.
(m) Timing device - A timing device accurate to plus or minus 0.2 second. Not less than 2 timing devices required.
(n) Depth micrometer - A depth micrometer capable of determining the depth of filter cake in the filter cell to an accuracy of 0.03 inch.
(o) Oven - A drying oven capable of obtaining 103° C.
(p) Beaker - A beaker having a capacity of 250 ml.
(q) Graph paper - Arithmetic graph paper, 10 by 10 to the centimeter.
(r) Distilled water - Fresh, free of bacteria and organics, and filtered through a filter membrane having a pore size of 0.45 micron.
(s) Inside micrometer - Capable of measuring diameter of .40 and .60 inch in thousands of an inch.

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4.4.4.2 Test procedures.

4.4.4.2.1 Effective specific gravity. The effective specific gravity (ℱF) shall be determined as follows:

(a) Determine the weight of a clean, dry 500 ml volumetric flask containing a magnetic stirring bar, recording the weight to the nearest 0.01 gm. This weight is the tare weight of the volumetric flask, W_T.
(b) Fill the volumetric flask three-fourths full with the sample of filter aid whose specific gravity is to be determined. Clean the outside of the flask and weigh the flask plus the filter aid plus the magnetic stirring bar, recording the weight, W_{TS}, to the nearest 0.01 gm.
(c) Slowly add distilled water to the flask to wet the filter aid thoroughly and to suspend it in a slurry, while stirring the flask contents with a magnetic stirrer. The distilled water shall be drawn from storage maintained at a temperature of 20° C to 25° C to reduce variation in water temperature during the test and to eliminate the need for temperature correction. Water shall be added until the flask is about half full of water. The removal of gases can be expedited by gently stirring the suspension in the flask with the magnetic stirrer. Then, the sample shall be allowed to soak for 30 minutes while stirring.
(d) After soaking, slowly add distilled water to the flask while the contents are being mixed to maintain a uniform suspension. When the water rises in the neck of the flask, quickly shut off the stirrer and bring the water level to the flask capacity graduation mark. Floaters in the filter aid may rise into the neck of the flask, obscuring the capacity mark. (Some practice may be required to add the proper amount of distilled water to reach this mark quickly, especially with perlite filter aids, before the floaters obscure the capacity mark.) Dry the outside of the flask and weigh the flask plus the magnetic stirring bar plus filter aid sample plus distilled water, recording the weight W_{TSW}, to the nearest 0.01 gm.
(e) Record the temperature of the contents of the flask to the nearest degree Celsius.
(f) Dispose of the contents of the flask, clean and replace the same magnetic stirring bar in the flask, and refill the flask with distilled water to the capacity graduation mark. The temperature of the distilled water shall be within plus or minus 1° C of the water used in step (c) and recorded in step (e).

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Weigh the flask plus the magnetic stirring bar plus the distilled water and record this weight, W_{TW}, to the nearest 0.01 gm.

(g) Record the temperature of the contents of the flask to the nearest degree Celsius.

(h) Dispose of the contents of the flask, clean and dry the flask.

(i) Record data on data sheet conforming to Figure 5.

4.4.4.2.2 Bulk density and resistance index. The bulk density (γF) and resistance index (β_o) shall be determined as follows:

(a) Prepare a filter aid suspension by adding to approximately 1100 ml of distilled water at about 20° C the weight of filter aid required to produce a cake 2.5 inches, plus or minus 0.2 inch, deep in the filter cell. A trial run using the following procedures should be made using a known weight of filter aid to be tested to determine weight of filter aid necessary to produce the required cake thickness. The suspension shall be left standing, with occasional or continuous mixing, for not less than 30 minutes to reach chemical equilibrium.

(b) Install a clean filter cake support (pad and filter membrane) in the filter cell and connect the filter cell to the sample holder (see Figure 3). Adjust the air pressure regulator to provide a pressure of 30 psi in the filter aid sample holder. Shut off the air pressure and add filter aid suspension to the filter aid sample holder. Keep suspension under continuous agitation using the magnetic stirrer and bar adjusted to 100 to 150 rpm. Allow the suspension to flow into the filter cell until almost all of the air in the filter cell is bled out by depressing the air-bleed valve. (If all of the air is bled out, some of the filter aid may lodge in the valve and cause leakage.)

(c) Start the filtration cycle by providing 30 psi of pressure to the system. The suspension will flow through the filter cell and form a cake on the surface of the filter cake support. Collect all of the filtrate in the graduated cylinder. When 100 ml of filtrate has been collected, one timing device shall be started. When 200 ml of filtrate has been collected, the timing device shall be stopped and another timing device simultaneously started. Repeat this procedure on successive 100 ml volumes of filtrate. Record the time for successive 100 ml volumes. Measure the temperature of the filtrate.

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Presence of filter aid in filtrate indicates poor sealing of membrane. The test shall be stopped, equipment checked, and test restarted from beginning.

(d) Leave the system under pressure, upon conclusion of timing of filtrate volume, until all possible filtrate has been forced through the filter cake (not less than 3 minutes). Shut off air pressure to filter cell, open air bleed valve to reduce pressure in filter cell and then disconnect the filter cell. Be careful not to disturb the cake. Measure the thickness of cake at not less than 3 places. Record the average thickness.

(e) Carefully remove the filter cake from the filter cell and collect in a tared beaker. Dry cake in drying oven at 130° C, plus or minus 2° C, to a constant weight.

(f) Record all data on data sheet conforming to Figure 6.

4.4.4.3 Calculations. For purposes of calculations, the viscosity of water shall be as stated in Table I.

TABLE I. Viscosity of water (μ).

Water temperature ° C	$\frac{\text{lb-sec}}{\text{ft}^2}$
10	2.7312×10^{-5}
11	2.6552 X
12	2.5821 X
13	2.5121 X
14	2.4455 X
15	2.3818 X
16	2.3206 X
17	2.2615 X
18	2.2053 X
19	2.1510 X
20	2.0990 X
21	2.0489 X
22	2.0006 X
23	1.9545 X
24	1.9094 X
25	1.8665 X
26	1.8248 X
27	1.7847 X
28	1.7460 X

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TABLE I. Viscosity of water (μ). - Continued

Water temperature ° C	$\frac{lb\text{-}sec}{ft^2}$
29	1.7084×10^{-5}
30	1.6723 X
31	1.6374 X
32	1.6038 X
33	1.5712 X
34	1.5395 X
35	1.5090 X
36	1.4797 X
37	1.4509 X
38	1.4231 X
39	1.3962 X
40	1.3701 X

4.4.4.3.1 Effective specific gravity. The effective specific gravity (ρF) shall be determined in accordance with Figure 5.

4.4.4.3.2 Bulk density of filter aid cake. The bulk density (γF) shall be determined as follows:

$$\gamma F = 3.81 \frac{W_c}{A \times T_c}$$

where γF = bulk density of filter aid cake (lb/ft^3)
W_c = weight of sample of filter aid (gm)
A = Area of filter cell (in^2)
T_c = Thickness of filter aid cake (in)

Area of filter cell shall be determined by inside micrometer.

4.4.4.3.3 Resistance index. The resistance index ($\beta\tau$) shall be determined as follows:

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$$\beta_o = 5.54 \frac{K_2 A^2 \Delta P V_T}{\mu W_c}$$

where β_o = resistance index (ft^2)
K_2 = The slope of a line made by plotting $\Delta T / \Delta V$ as ordinate versus V as abscissa on graph paper and drawing best straight line (sec/ml^2)
ΔP = Pressure of system (lb/in^2)
μ = viscosity of water ($lb\text{-}sec/ft^2$)
W_c = weight of sample (gm)
V_T = Volume of filtrate (ml)

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Quantities per sack not as specified.
102.	Shipping sacks not as specified.
103.	Filter aids not unitized as specified for Level A.
104.	Filter aids not palletized as specified for Level B.
105.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation (see 6.6). Filter aids, in quantities of 3.0 cubic feet (plus or minus 0.1 cubic foot), shall be contained in paper shipping sacks conforming to UU-S-48, Type IV, sack number 8 or 8X, MB Grade 1 (see 6.5).

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

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5.2.1 Level A. Filter aids, contained in shipping sacks specified in 5.1, shall be unitized in accordance with requirements of Figure 7, Level A packing.

5.2.2 Level B. Filter aids, contained in shipping sacks specified in 5.1, shall be palletized in accordance with requirements of MIL-STD-147, Load type XV.

5.2.3 Commercial. Filter aids, contained in shipping sacks specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking (see 6.6).

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The filter aid is intended for use in military water purification equipment for filtering potable water.

6.2 Ordering data. Procurememt documents should specify the following:

(a) Title, number, and date of this specification.
(b) Test mediums required if other than specified (see 4.4.1).
(c) Degree of packing required (see 5.2).

6.3 Items required for determination of index value may be obtained from the following.

Flask - E.H. Sargent and Co., Flask 534855-20, Polyethylene Stopper, Class A, Pyrex Glass, Size K (500 ml).
Balance - E.H. Sargent and Co., Balance S-2552-40, top-loading, direct-reading, one-pan, 1200 gm, Mettler P-1200.
Thermometer - E.H. Sargent and Co., Thermometer S-80006B, specific gravity, ASTM No. E 1 (12C), centigrade.
Stirring apparatus - E.H. Sargent and Co., Stirring apparatus S-76490, magnetic, variable-speed.
Wash bottle - E.H. Sargent and Co., Bottle-washing, S9487C, Polyethylene, One Piece Cap, Size C (500 ml).

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Compressor - W.W. Granger, 1/2 HP, 12-gallon portable utility compressor, 1.86 CFM free air at 60 psi, No. 1Z910.
Air-pressure regulator - Fisher Governor Co., Type 67 air-pressure regulator, maximum inlet - 250 psi, outlet range 0 to psi, plus or minus 0.1 psi.
Manometer - Meriam Instrument Co., manometer range - 100 inches, Model 30EB25 WM, stainless steel construction, scale (psi and tenths),.
Pressure gage - Marshalltown Manufacturing Inc., inspectors test gage, Figure No. 63, 0-30 psi, 1/4 psi marking divisions, accuracy - plus or minus 0.5 percent of full scale reading.
Filter membranes - Millipore Filter Co., MF-Millipore membranes, Type HA, 0.45 micron, white, plain, 37 mm.
Collection vessel - E.H. Sargent and Co., S-24665J cylinder, single, graduated, Pyrex brand glass, 1000 ml.
Timing device - E.H. Sargent and Co., S-77446 stopwatch, 60 minutes, 0.2 second, single continuous, decimal, precision grade.
Absorbent pads - Millipore Filter Co., AP-1003700, 37 mm.
Depth micrometer - Iowa Machinery and Supply Co., Lufkin depth gage, No. H511A.
Oven - E.H. Sargent and Co., S-64055 oven, Thelco No. 14, gravity convection, electric, 150° C, single wall.
Beaker - E.H. Sargent and Co., S-4675K beaker, Griffin, low form, with pourout, graduated, Pyrex glass, 250 ml.
Graph paper - Kueffel and Esser Co., 10 by 10 to the centimeter, 461510.
Micrometer - The L.S. Starrett Co., 700A, .200 to 1.200 inch range, thousandths of an inch graduations, with lock screw.

6.4 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform (see 4.3.1).

6.5 Packaging density. For purposes of determining cubic feet of filter aid per bag, the packaging density (lbs/ft^3) used by the contractor for Commercial packaging shall be used. Cubic feet of filter aid in the bag is equal to the weight of the bag in pounds (lbs) divided by the density in lbs/ft^3.

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6.6 Level of preservation (see 5.1). The items procured under this specification are for use in forward areas. The nature of the items themselves along with their intended use is such that the preservation specified is the minimum acceptable for any level of protection. For purposes of preservation/packing level marking, the preservation shall be considered Level A.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MD
DLA - GS

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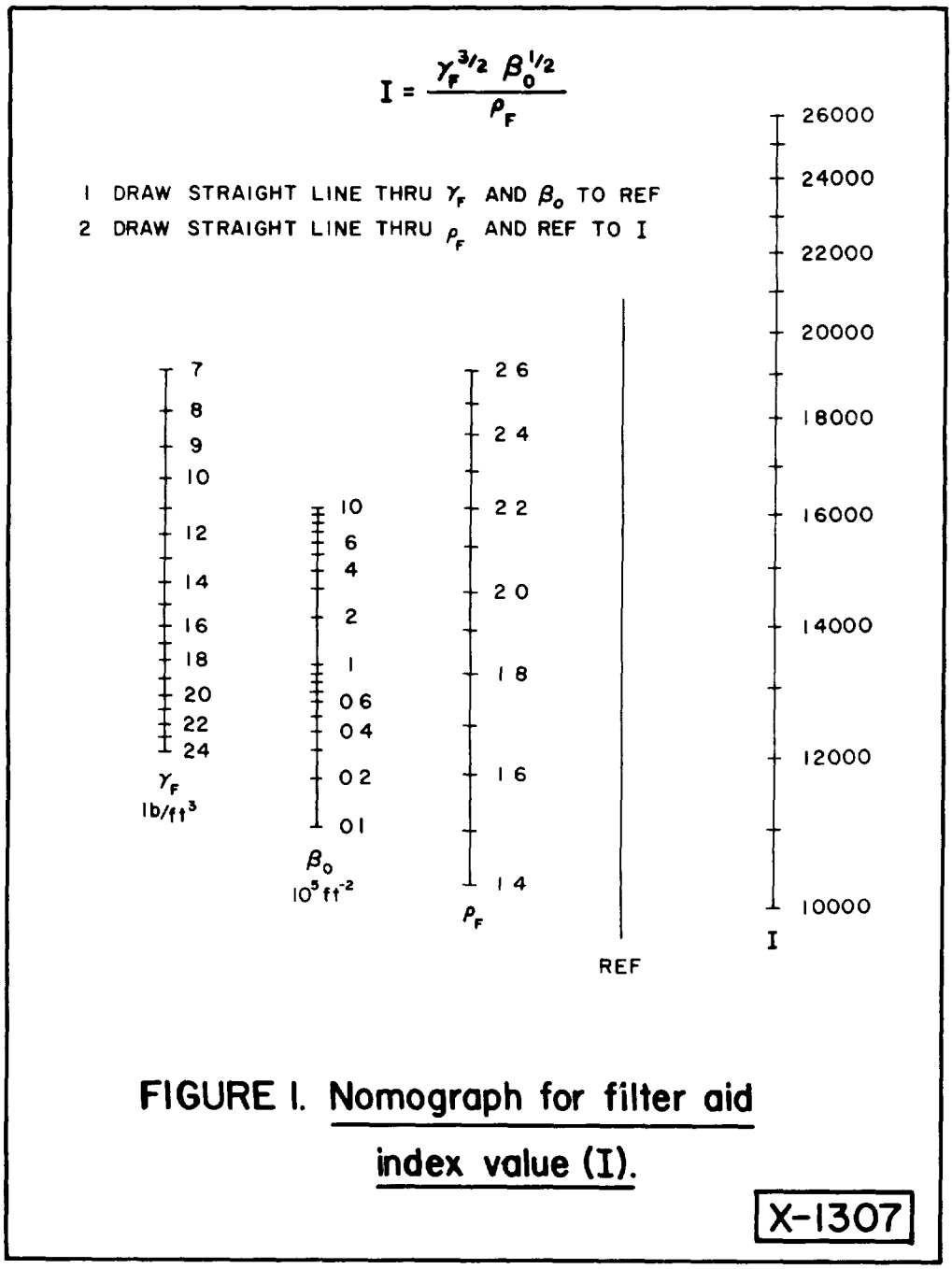


FIGURE I. Nomograph for filter aid index value (I).

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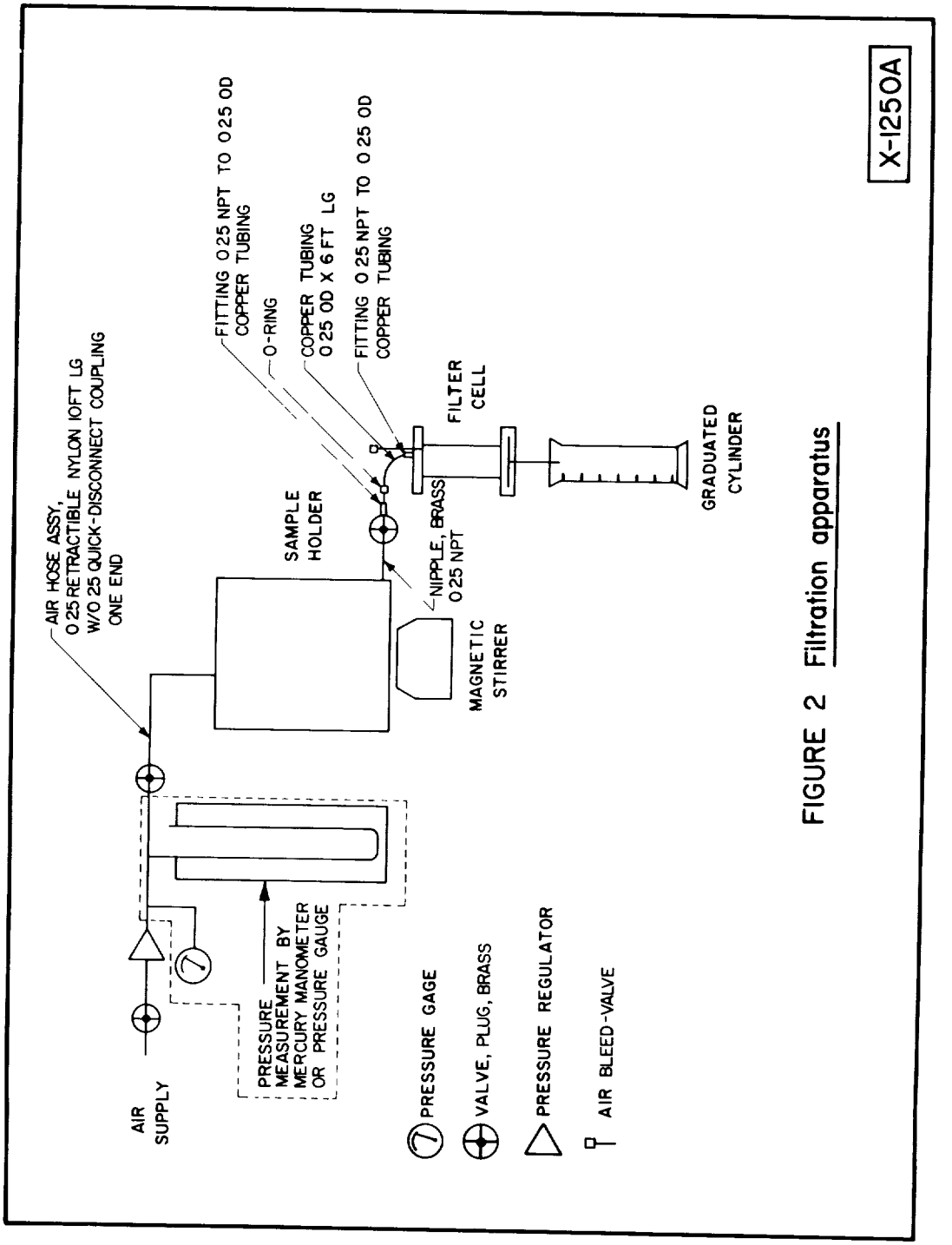


FIGURE 2 Filtration apparatus

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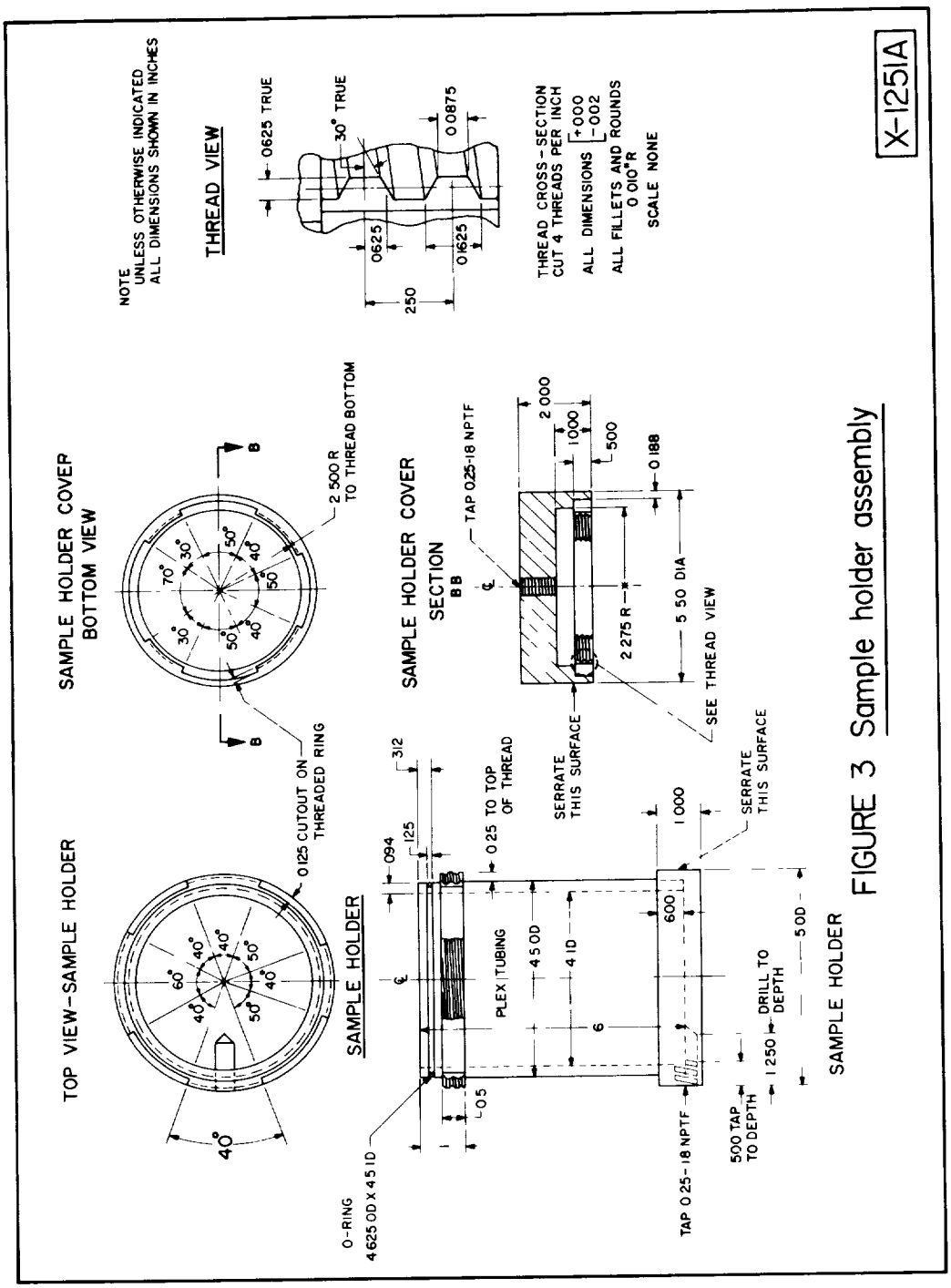


FIGURE 3 Sample holder assembly

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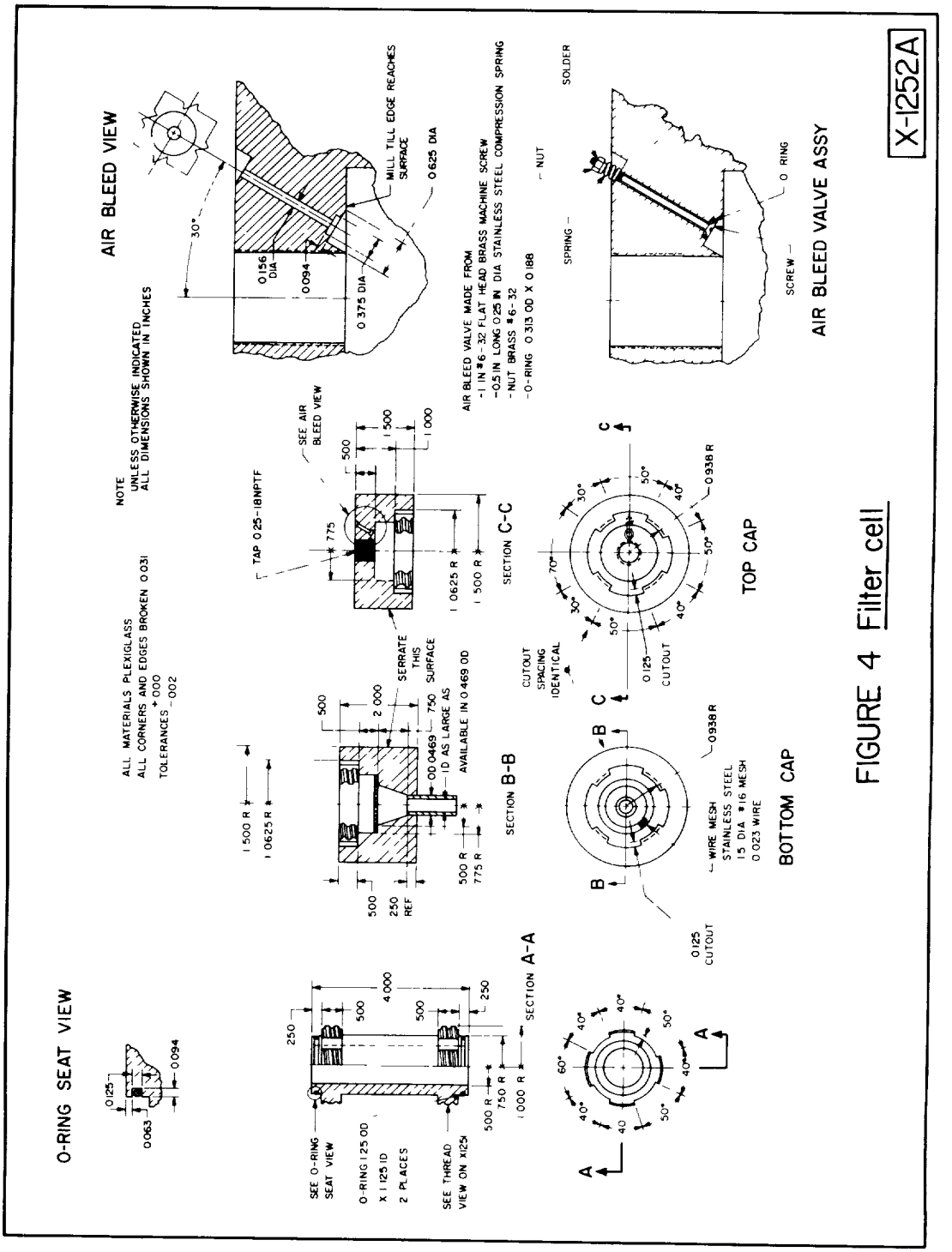


FIGURE 4 Filter cell

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Data Sheet

EFFECTIVE SPECIFIC GRAVITY DETERMINATION

FOR

FILTER AIDS

Sample designation:_____ Date:_____

Volumetric flask number:_____ Technician:_____

Temperature of water when W_{TW} is determined_____ °C

Weight of volumetric flask plus sample, W_{TS} = _____ gm

Weight of volumetric flask, W_T = _____ gm

Weight of filter aid sample, $(W_{TS} - W_T)$ = _____ gm

Temperature of water when W_{TSW} is determined = _____ °C

Weight of volumetric flask plus sample plus water, W_{TSW} = _____ gm

Weight of volumetric flask plus water, W_{TW} = _____ gm

$(W_{TSW} - W_{TW})$ = _____ gm

$$\rho F = \frac{\text{weight of sample}}{\text{weight of water displaced}} = \frac{(W_{TS} - W_T)}{(W_{TS} - W_T) - (W_{TSW} - W_{TW})}$$

= _____

ρF = effective specific gravity = _____

1. All weights shall include the magnetic stirring bar in the flask.
2. All calculations shall be made on a separate sheet and attached to this sheet.

FIGURE 5. Data sheet for effective specific gravity (ρF).

X-3345

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Data Sheet

DETERMINATION OF BULK DENSITY, FILTRATION RESISTANCE AND FILTER AID INDEX

Series_____Run No._____Suspension._____Operator:_____Date _____

Pressure, Δ P(psi) _____

Filtrate Temperature (°C)_____

Filter Area, A(in.^{2})_____

Viscosity, μ ($\frac{\text{lb-sec}}{\text{ft}^2}$)_____

TEST DATA

V ml	ΔV ml	ΔT sec	ΔT/ΔV sec/ml
100	100	_____	_____
200	100	_____	_____
300	100	_____	_____
400	100	_____	_____
500	100	_____	_____
600	100	_____	_____
700	100	_____	_____
800	100	_____	_____
900	100	_____	_____
1000	100	_____	_____

Height of Filter Cake, T_c (in.) _____
Volume of Filterate, V_T (ml) _____
Weight of Beaker and Sample (gm) _____
Weight of Beaker No. _____ (gm) _____
Weight of Sample, W_c (gm) _____

CALCULATIONS

Bulk Density, γF (lb/ft^3) = $3.81 \frac{W_c}{AT_c}$ = _____

Filtration Resistance Factor, K_2 (sec/ml^2)
(from plot of ΔT/ΔV vs V) = _____

Filter Cake Resistance, β_o (ft^2) = $5.54 \frac{K_2 A^2 \Delta P V_T}{\mu W_c}$ = _____

Filter Aid Index, I = $\frac{\gamma F^{3/2} \beta_o^{1/2}}{\rho F}$ = _____
(from nomograph)

All calculations shall be made on a separate sheet and attached to this sheet.

FIGURE 6. Data sheet for filter aid index (I).

X-3346

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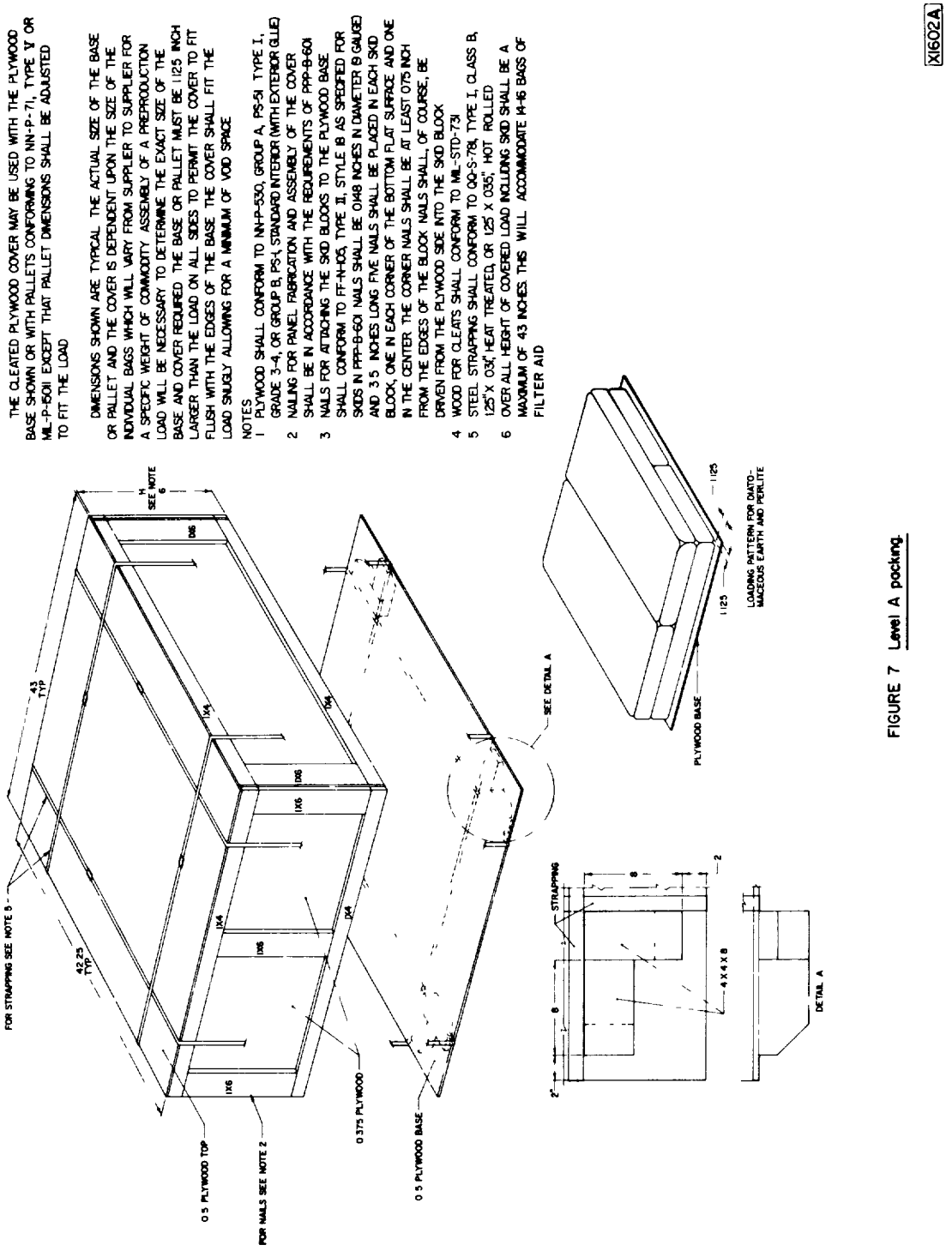


FIGURE 7 Level A packing

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MIL-F-52637A(ME) Filter Aids, Water Purification

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